

TEMPLATE

KEY PROJECT INFORMATION & VPA DESIGN DOCUMENT (PDD)

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VERSION **v. 1.1**

RELATED SUPPORT

- **TEMPLATE GUIDE Key Project Information & VPA Design Document v.1.1**

This document contains the following Sections

Key Project Information

Section A – Description of project

Section B - Application of approved Gold Standard Methodology (ies) and/or demonstration of SDG Contributions

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Section D – Summary of Safeguarding Principles and Gender Sensitive Assessment

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Q - Summary of Approved Design Changes (project specific)

KEY PROJECT INFORMATION

GS ID of Project	11143
Title of Project	GS 11142- Installation of High Efficiency Firewood Cookstoves in Sub-Saharan Africa Installation of High Efficiency Firewood Cookstoves in Zambia- VPA1
Time of First Submission Date	18/03/2021
Date of Design Certification	NA
Version number of the VPA-DD	3.0
Completion date of version	21/04/2022
Coordinating/managing entity	C-Quest Capital Stoves Asia
VPA Implementer (s)	NA
Project Participants and any communities involved	NA
Host Country (ies)	Republic of Zambia
GS ID and Title of applicable Design Certified VPA	NA
GS ID and Title of applicable Performance Certified VPA	NA
Activity Requirements applied	☒ Community Services Activities ☐ Renewable Energy Activities ☐ Land Use and Forestry Activities/Risks & Capacities ☐ N/A
Scale of the project activity	☐ Micro scale ☐ Small Scale ☒ Large Scale
Other Requirements applied	Usage Rate Requirements
Methodology (ies) applied and version number	TPDDTEC (V3.1)

Product Requirements applied	☒ GHG Emissions Reduction & Sequestration ☐ Renewable Energy Label ☐ N/A
Project Cycle:	☒ Regular ☐ Retroactive

Table 1 – Estimated Sustainable Development Contributions

Sustainable Development Goals Targeted	SDG Impact (defined in B.6Error! Reference source not found.)	Estimated Annual Average	Units or Products
13 Climate Action (mandatory)	Emission reductions	254,166 t CO ₂ /year	
15 Life on Land	Amount of firewood saved by all the project stoves per year	171,807 tons/year	
3 Good health and well-being	Proportion of households who perceive reduced indoor air pollution	100%	
5 Gender Equality	Proportion of households who perceive reduced time for collecting wood and cooking	100%	
5 Gender Equality	Number of women artisans who received training	20 number	
8 Decent Work and Economic Growth	Number of temporary and permanent jobs created	50 number	
1 No Poverty	Increased access to basic services	50000 number	

SECTION A. DESCRIPTION OF PROJECT

A.1. Purpose and general description of project

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C-Quest Capital plans to install 50,000 domestic high efficiency cookstoves in total in rural Zambia where 84.5% of the households rely on firewood as main source of energy for cooking¹. Fuelwood consumption has been increasing during last two decades and has doubled from 11.3 millionm³ in 1990 to 23 million m³ in 2018². On the other hand, forest cover in Zambia has been constantly decreasing and 2.6 million ha of forest land has been lost in last 30 years. The rate of deforestation has accelerated between 2015-2020. 1.1 million ha of total forest has been lost with an average rate of 188 thousand ha annually (0.4% of the total forest area) in the same period (Global Forest Resources Assessment (FRA) 2020- Zambia Report³).

The project stoves disseminated will replace the prevailing inefficient three stone fires or traditional pot supports with stoves which combust wood more efficiently and improve thermal transfer to pots, hence saving fuel and lowering greenhouse gas emissions. The ICL envisaged to be implemented under the VPA is TLC Rocket Stove, which results in substantially reduced fuel consumption and emissions for conducting cooking and water heating tasks in homes.

The project plans to install improved cookstoves to the households in rural regions of Zambia and Gold Standard approved Community Services Activity Requirements is applicable for the project. The project type is, therefore; automatically eligible as per section 4.1.3 of GS4GG Principles & Requirements.

Republic of Zambia is listed as Least Developed Country by UN and does not have an emission cap enforced. There is no policies enforced/under development for trading emission under the scope of the proposed project.

The project is compliance with all applicable legal, environmental, ecological and social regulations in Zambia. Improved cooking devices to include improved biomass stove,

¹ https://www.zamstats.gov.zm/phocadownload/Living_Conditions/2015%20Living%20Conditions%20Monitoring%20Survey%20Report.pdf

² <http://data.un.org/Data.aspx?d=EDATA&f=cmID%3aFW%3btrID%3a12>

³ <http://www.fao.org/3/cb0092en/cb0092en.pdf>

use of ethanol and LPG stoves, and switch to electric stoves is considered as a mitigation measure under the Sustainable Forest Management programme in Intended Nationally Determined Contributions ⁴. The implementation is said to be subject to the availability of international support in form of finance, technology and capacity building; which is now provided by carbon finance.

A.1.1. Eligibility of the project under approved PoA

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No.	Eligibility Criterion	Description/ Required condition	Means of Verification/Supporting evidence for inclusion
1	Geographic boundary	Each VPA shall be located within the boundaries of the PoA.	Recorded geographical location in the project database to be checked by VVB
2	Double Counting	A unique numbering system will be adopted for each VPA. All ICS installed shall receive a unique number identifying to which VPA it belongs.	Each ICS included in the VPA will have a unique combination of customer name and geographical location linked with a unique serial number. The Project Developer commits to retiring eligible units equal to the quantity of GS VERs if any double counting exists. For instances where stoves are transferred from CME's managed other GHG programme to VPAs under this PoA, a zero-issuance certificate from the GHG programme registry for the transferred stoves shall be submitted to verifying VVB at the time of each verification.
3	Exclusiveness of VPA	VPA shall not be previously: 1. Registered as a project activity with	Confirmation by CME

⁴ https://www4.unfccc.int/sites/ndcstaging/PublishedDocuments/Zambia%20First/FINAL+ZAMBIA%27S+INDC_1.pdf

		other offset schemes 2. Included as a VPA in any other registered PoA, or deregistered as a VPA of a PoA.	
4	Specification of Technology/ Measure	VPAs under the PoA shall install in-situ ICS using firewood. The capacity is limited by 150kW as per the applied TPDDTEC methodology.	Each stove is assumed to have a thermal capacity of 2.78 kW. The real capacity will be determined by Kitchen Performance Test (KPT) and Water Boiling Test (WBT) results
5	Start date	The start date of any proposed VPA will be the date of installation of first device under it.	The date on which first ICS installed will be recorded. Sales invoice or end-user agreement will be provided as a proof.
6	Applicability of methodologies	VPAs shall apply TPDDTEC (v 3.1).	Please refer to Section B.2 of VPA-DD for confirmation.
7	Additionality	The additionality PoA is demonstrated as per GS4GG Community services activity requirements, Version 1.2	• Confirmation by the CME that the project is located in Zambia to be provided to the VVB. • Total thermal energy savings per unit is 0.017 GWh. Actual values shall be submitted to VVB at the time of verification.
8	Public funding	Affirmation that there is no diversion of Official Development Assistance (ODA).	Declaration of CME
9	Target group & Distribution Mechanism	Households will be the target group for VPAs. ICSs will be installed by local VPA Implementers/CME representatives on a non-commercial basis.	CME declaration or user registration.
10	Sampling requirements	All requirements as mentioned in TPDDEC, version 3.1 or the Standard: Sampling and surveys for CDM project activities and programme of activities are applicable to VPAs.	Please see Section B.7.2 Sampling Plan for details. VPA Implementers will follow the management system described in the PoA-DD.

11	Compliance of the technology implemented	The capacity of each ICS will comply with the requirements of TPDDTEC methodology.	VPA applies TPDDTEC (v3.1) Please refer to Section B.2 of the VPA-DD for demonstration of its compliance with TPDDTEC requirements.
12	SDG Outcomes	Each VPA shall conduct SDG outcomes assessment and comply with the SDG targets identified in the PoA-DD.	VPA-DD includes SDG outcomes assessment as defined in the PoA. Please refer to Section B.6 of the VPA-DD.
13	Stakeholder Consultation	Each VPA will conduct a local stakeholders' consultation.	Local Stakeholder Consultation Meeting was conducted on December 10 th 2020. Please see Stakeholder Consultation Report for details.
14	VER Ownership	End users receiving ICSs under the specific VPA contractually cede their rights to claim and own emission reductions to the CME of the PoA.	A sample carbon waiver document has been provided to VVB.

A.1.2. Legal ownership of products generated by the project and legal rights to alter use of resources required to service the project

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The Verified Carbon Reductions (VER) generated by each stove belongs to the CME. The stove registration document signed by the stove user, includes terms transferring the ownership of VERs from user to the CME. By agreeing to participate in the program, the users accept to waive their carbon rights

A.2. Location of project

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Republic of Zambia



Figure 1: Map of Zambia

Coordinates for Zambia: 13.1339° S, 27.8493° E

A.3. Technologies and/or measures

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The TLC Rocket Stove is a simple design with basic features. The design uses a total of 15 readily available building bricks that are made by the household using locally available clay. The average size of the brick used on the TLC Rocket Stove which is produced using a standard mould is 22cm (L) x 11cm (W) x 6.5cm (H). The bricks are mortared together using locally available material (clay soil, cow dung, and sand) for better insulation and heat loss reduction. The mud mortar is a mix of 5 liters clay, 5 liters sand, 5 liters manure with 5 liters of water. Average lifespan for the cookstoves is expected to be 10 years. Figure 1. below shows how the cook stove is assembled.

Table 2. Technical Specifications

Technical Specifications – TLC Rocket Stove	
Thermal efficiency	34.5 % ⁵
Size (stove)	Depth: 34.5 cm

⁵ Based on the WBT results of TLC 15 Brick Rocket Stove, October 2015, Aprovecho Research Center

	Width: 34.5 cm Height: 28 cm
Size (combustion chamber)	Depth: 12 cm Width: 12 cm Height: 28 cm

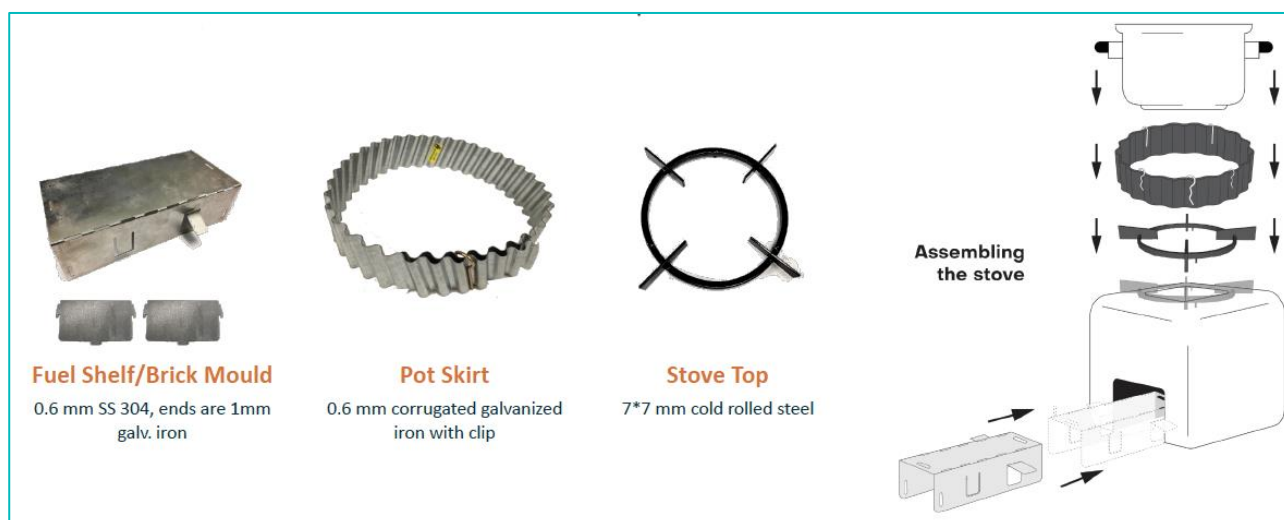


Figure 1. Parts of the project stove

The stainless-steel stick shelf allows air to flow underneath the wood sticks into the combustion chamber resulting in cleaner and more efficient combustion.

The adjustable corrugated steel pot skirt improves the transfer of heat from the fire into the pot, increasing overall energy efficiency of the stove.

A.4. Scale of the project

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The project is large scale based on project scale defined under Section 9.1.2(b) in GHG Emissions Reduction & Sequestration Product Requirements (v2.0). The aggregate energy savings of the project activity exceeds the limit of small scale, equivalent of 180 GWh thermal per year in fuel input. Each ICS installed saves 0.017 GWh and 50,000 stoves are planned to be distributed, which is 850 GWh thermal energy savings per year. Thermal efficiency savings per stove is calculated as follows:

$$TE_{savings} = B_{old} \times \left(1 - \frac{n_{old}}{n_{new}}\right) \times NCV_{biomass} \times f_{GWh}$$

$$TE_{savings} = 0.0148 \frac{\text{ton}}{\text{day}} \times \left(1 - \frac{0.1}{0.345}\right) \times 365 \frac{\text{days}}{\text{year}} \times 0.0156 \frac{\text{TJ}}{\text{ton}} \times 0.27778 = 0.017 \text{ GWh}$$

- $TE_{savings}$: Thermal Energy savings per project stove (GWh)
 B_{old} : Annual fuel consumption in baseline scenario per unit per year (ton/y)
 n_{old} : Default value for thermal efficiency of the three stone baseline stove (%)
 n_{new} : Thermal efficiency of the project stove as per WBT results (%)
 $NCV_{biomass}$: Net calorific value for firewood (TJ/ton)
 f_{GWh} : Conversion factor for TJ to GWh

A.5. Funding sources of project

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The project will be implemented by C-Quest Capital; the producer of the improved cookstoves and Project Developer, in cooperation with the local partners or Project Implementers. The funding for the production of stoves will be provided by C-Quest Capital which will be legal owner of the VERs. No public funding or Official Development Assistance is involved. The ODA Declaration signed by the CME will be provided as per GS4GG requirements.

SECTION B. APPLICATION OF APPROVED GOLD STANDARD METHODOLOGY (IES) AND/OR DEMONSTRATION OF SDG CONTRIBUTIONS

B.1. Reference of approved methodology (ies)

>>GS Methodology: Technologies and Practices to Displace Decentralized Thermal Energy Consumption” (TPDDTEC) Version 3.1 (August 2017)

Applicable GS4GG requirements are as follows:

- Principles and Requirements (V1.2)
- GHG Emissions Reduction & Sequestration Product Requirements (v2.0)
- Community Services Activity Requirements (v1.2)
- Usage Rate Requirements (27/10/2020)

B.2. Applicability of methodology (ies)

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TPDDTEC (V3.1) methodology defines the following applicability criteria:

Applicability	Justification
The project boundary needs to be clearly identified, and the technologies counted in the project are not included in any other voluntary market or CDM project activity (i.e. no double counting takes place). In some cases, there may be another similar activity within the same target area. Project proponents must therefore have a survey mechanism in place together with appropriate mitigation measures so as to prevent any possibility of double counting.	The project boundary will be within the territorial limits of Zambia. Each stove installed under the VPA shall be identified through a combination of unique serial number and GPS coordinates enabling their identification with the VPA. Prior to each verification, the project database will be checked to make sure that only stoves belonging to the VPA are included for emission reduction calculation as well as stoves included in this VPA are not included in any other GHG program or VPAs belonging to this particular PoA. The registration card with corresponding stove serial number will ensure that no double counting of stoves takes place. If stoves are transferred from other GHG programme to the said VPA, a zero-

	issuance certificate from the programme registry shall be submitted to the verifying VVB to ensure that credits are not being claimed under GS and other programme for same vintage.
The technologies each have continuous useful energy outputs of less than 150kW per unit (defined as the total useful energy delivered from start to end of operation of a unit divided by time of operation). For technologies or practices that do not deliver thermal energy in the project scenario but only displace thermal energy supplied in the baseline scenario, the 150kW threshold applies to the displaced baseline technology.	The capacity of each stove is calculated as 2.78kW to achieve the calculated energy savings ⁶ . That is only 1.88% of the 150kW limit. Please see the calculation below.
Using the baseline technology as a backup or auxiliary technology in parallel with the improved technology introduced by the project activity is permitted as long as a mechanism is put into place to encourage the removal of the old technology (e.g. discounted price for the improved technology) and the definitive discontinuity of its use. The project documentation must provide a clear description of the approach chosen and the monitoring plan must allow for a good understanding of the extent to which the baseline technology is still in	Continued use of baseline stoves will be monitored during annual/biennial surveys and the emissions related with the use will be accounted in the overall emission reduction amount.

⁶ The value is based on the WBT results of TLC 15 Brick Rocket Stove, October 2015, Aprovecho Research Center. The real value will be determined by the results of field performance tests (WBT and KPT).

use after the introduction of the improved technology.	
The project proponent must clearly communicate to all project participants the entity that is claiming ownership rights of and selling the emission reductions resulting from the project activity. For technology producers and the retailers of the improved technology or the renewable fuel in use, this must be communicated by contract or clear written assertions in the transaction paperwork. If the claimants are not the project technology end users, the end users will need to be informed and notified that they cannot claim for emission reductions from the project	Each user will be informed about the transfer of carbon rights and an agreement will be signed between Project Developer and the user. How the project generates carbon credits and what kind of project activities will be financed by the revenue from carbon credits was discussed in detail during the Stakeholder Consultation Meeting as well.
Project activities making use of a new biomass feedstock in the project situation (e.g. shift from non-renewable to green charcoal, plant oil or renewable biomass briquettes) must comply with relevant Gold Standard specific requirements for biomass related project activities, as defined in the latest version of the Gold Standard rules	N/A

Thermal capacity of the project stove is calculated as follows:

Step 1: Estimate the energy consumption in kWh

$$B_{new} = B_{old,i,j} \times \left(\frac{n_{old}}{n_{new}} \right) = 0.0148 \frac{\text{ton}}{\text{day}} \times \left(\frac{0.1}{0.345} \right) = 0.0043 \frac{\text{ton}}{\text{day}}$$

$$TE = 0.0043 \frac{\text{ton}}{\text{d}} \times 0.0156 \frac{\text{TJ}}{\text{ton}} \times 277,778 \frac{\text{kWh}}{\text{TJ}} = 18.6 \text{ kWh/d}$$

Step 2: Estimate energy output:

Based on a survey done by World Bank for Sub-Saharan countries, households spend 3 hours on average for cooking⁷. Accounting that 77%⁸ of cooking is done with the project stove, number of hours for the project stove is calculated as 2.31 hours in a day.

Therefore, the capacity of stove is $18.6 \text{ kWh}/2.31\text{h} = 8.0 \text{ kW}$.

Step 3: Estimate the useful thermal energy output:

Assuming the thermal efficiency of the stove is 34.5%; the useful output will be $8.0 \text{ kW} \times 0.345 = 2.78 \text{ kW}$.

B.3. Project boundary

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The Project boundary is physical, geographical site of individual improved cookstoves that utilize biomass.

Source		GHGs	Included?	Justification/Explanation
Baseline scenario	Source 1: Combustion of non-renewable biomass for cooking on traditional cookstoves	CO ₂	Yes	Major source of emissions
		CH ₄	Yes	Important source of emissions
		N ₂ O	Yes	Can be significant source of emissions for some fuels
Project scenario	Source 1: Combustion of nonrenewable biomass for cooking with ICS	CO ₂	Yes	Major source of emissions
		CH ₄	Yes	Important source of emissions
		N ₂ O	Yes	Can be significant source of emissions for some fuels

⁷

<https://openknowledge.worldbank.org/bitstream/handle/10986/22521/Clean0and0impr000a0landscape0report.pdf?sequence=1&isAllowed=y> (Page:154)

⁸Monitoring report dated 13/08/3019 for CPAs of CDM PoA8060 states that 23% of the households uses a second stove at least once per week as per the baseline study conducted.

B.4. Establishment and description of baseline scenario

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As per the applied methodology, the project proponent must conduct the following baseline studies for each baseline scenario:

- Baseline non-renewable biomass (NRB) assessment, if biomass is one of the baseline fuels
- Baseline survey (BS) of target population characteristics
- Baseline performance field test (BFT) of fuel consumption (e.g. kitchen performance test (KPT) in case of cook stoves)

Baseline non-renewable biomass (NRB) assessment has been calculated by an independent study carried out as per "CDM Tool 30: calculation of the fraction of non-renewal biomass"; version 3.0 by third party.. The fNRB value for Zambia is calculated to be 81.4% and will be used in emission reduction calculations.

Baseline survey (BS) of target population characteristics

The baseline survey provides critical information on target population characteristics, baseline technology use, fuel consumption, leakage, and sustainable development indicators.

Baseline Survey Representativeness

The baseline survey requires in person interviews with a robust sample of end users without project technologies that are representative of end users targeted in the project activity.

Rural households in Zambia using fuelwood for cooking are the target population and the following data were used.

Table 3. Characteristics of target population

Data Used	Value	Source
Population of Republic of Zambia (2018)	17,861,030	World Bank ⁹

⁹ <https://data.worldbank.org/indicator/SP.POP.TOTL?locations=ZM>

Share of population using wood for cooking	51.70%	Zambia Demographic and Health survey 2018 ¹⁰
Population using wood for cooking	9,234,153	Calculated
Average Household size	5	Zambia Demographic and health survey 2018
Total households using fuelwood	1,846,831	Calculated

Baseline Survey Sample Sizing

In accordance with applied methodology, the baseline survey has to be carried out for each baseline scenario using representative and random sampling, following these guidelines for minimum sample size:

- Group size <300: Minimum sample size 30 or population size, whichever is smaller
- Group size 300 to 1000: Minimum sample size 10% of group size
- Group size > 1000: Minimum sample size 100

The baseline survey was conducted in Kafue District of Lusaka Province between June 24th2021 and July, 08th 2021. 110 households were sampled in Lukolongo and Shimabala wards using simple random sampling.

Place	Number of Respondents	Female	Male
Lukolongo	85	75	10
Shimabala	25	25	0

The following information has been gathered:

1. User follow up
 - a. Address or location
 - b. Mobile telephone number and/or landline telephone number (where possible)
2. End user characteristics
 - a. Number of people served by baseline technology

¹⁰ <https://dhsprogram.com/publications/publication-fr361-dhs-final-reports.cfm>

3. Baseline technology and fuels

- a. Types of baseline technologies used and estimated frequency
- b. Types of fuels used
- c. Sources of fuels; (purchased or -collected, etc) and prices paid, or effort made (e.g. walking distances, persons collecting, opportunity cost)
- d. Renewability and non-renewability indicators as required by Annex 1

The survey included questions about basic household characteristics including the number of people served in the household, the number of meals cooked per day, stove and fuel use in the current baseline scenario and season variations of stove and fuel use patterns. Contact details of the participants are also recorded.

Results and analysis

- 94.5% of the households use three-stone fire and fuelwood as the main source of energy.

For those who use three stone fire and fuelwood:

- 85.6% of households cooks 3 meals per day and none of them has a secondary stove.
- 70% of the households who use three stone fire cook in a separate enclosed kitchen while 29% cooks outside, 1% cooks inside.
- 95% of them experience health problems.
- In 74% of total households the number of people served do not change between seasons.
- Except one respondent, all of the households collect fuelwood. 92% of them collect 100% their fuelwood.

For those who collect 100% of their fuelwood:

- They collect fuelwood around the village (31%), forest (48%) and own land (21%).
- In 91.6% of the households, female adult collect fuelwood.
- Most of them (92.7%) do 2 or 3 trips per week for the purpose, spending 106 minutes and 1.66 kms on average for each trip. Almost all of them confirmed that length of each trip is increasing by time.

Baseline performance field test (BFT) of fuel consumption

As per the methodology, the baseline Kitchen Performance Test (KPT) should include a sample of end-users without project technologies that are representative of the end users targeted in the project activity. The baseline survey and baseline KPT can be conducted concurrently with the same end users. Any sampling methods can be used.

In cases of paired and independent sampling, there are two valid options for the statistical analysis:

- 90/30 rule: When the sample size is large enough to satisfy the 90/30 rule, i.e. the endpoints of the 90% confidence interval lie within +/- 30% of the estimated mean, overall emission reductions can be calculated on the basis of the estimated MEAN annual emission reduction per unit or MEAN fuel annual savings per unit.
- 90% confidence rule (Lower bound of the one-sided 90% confidence interval). When the result obtained does not satisfy 90/30 precision, the emission or fuel saving result is not the mean (or average) test result, but a lower value, i.e. the LOWER BOUND of the one-sided 90% confidence interval.

Independent sampling for KPT – sampling of baseline stoves and project stoves are independent of each other. Among 110 end users in the baseline survey, 35 households with similar socioeconomic and demographic characteristics within the targeted population were selected to perform baseline KPTs. The households did not use the project stove or any other /ICS prior to the tests. With two outliers removed, the average quantity of fuelwood used per day is 14.8 kg/household/day. The survey results and calculation have been submitted as a separate excel file.

B.5. Demonstration of additionality

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Specify the methodology, activity requirement or product requirement that establishes deemed additionality for the proposed project (including the version number and the specific paragraph, if applicable).	Activity Requirement for Community based technologies: 4.1.9. Projects that meet any of the following criteria are considered as deemed additional and therefore are not required to prove Financial Additionality at the time of Design Certification: (a) Positive list (Annex B)
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	(b) Projects located in LDC, SIDS, LLDC (c) Micro-scale projects
Describe how the proposed project meets the criteria for deemed additionality.	The proposed project is located in the Republic of Zambia which falls under the category of a LDC and LLDC. The project also complies with the requirement 1.1.3 of Annex B: Project activities solely composed of isolated units where the users of the technology/measure are households or communities or institutions and where each unit results in ≤ 600 MWh (1.8 GWh_{th}) of energy savings per year or ≤ 600 tonnes of emission reductions per year. Each stove is assumed to be save 0.017 GWh and 5.44 tCO₂e per year.

B.5.1. Prior Consideration

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The project is applying under regular project cycle and demonstration of prior consideration is not required.

B.5.2. Ongoing Financial Need

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Not applicable

B.6. Sustainable Development Goals (SDG) outcomes

Relevant Target/Indicator for each of the three SDGs

Sustainable Development Goals Targeted	Most relevant SDG Target	SDG Impact
		Indicator (Proposed or SDG Indicator)

13 Climate Action (mandatory)	13.b. Promote mechanisms for raising capacity for effective climate change related planning and management in least developed countries and small island developing States, including focusing on women, youth and local and marginalized communities	Emission reductions achieved by fuelwood savings at household level
15 Life on Land	15.2. By 2020, promote the implementation of sustainable management of all types of forests, halt deforestation, restore degraded forests and substantially increase afforestation and reforestation globally.	Amount of firewood saved by the project stoves per year
3 Good health and well-being	3.9. By 2030, substantially reduce the number of deaths and illnesses from hazardous chemicals and air, water and soil pollution and contamination	Proportion of households who perceive reduced indoor air pollution
5 Gender Equality	5.4. Recognize and value unpaid care and domestic work through the provision of public services, infrastructure and social protection policies and the promotion of shared responsibility within the household and the family as nationally appropriate	Proportion of households who perceive reduced time for collecting wood and cooking Number of women artisans who received training
8 Decent Work and Economic Growth	8.5. By 2030, achieve full and productive employment and decent work for all women and men, including for young people and persons with disabilities, and equal pay for work of equal value	Number of temporary and permanent jobs created

1 No poverty	1.4 By 2030, ensure that all men and women, in particular the poor and the vulnerable, have equal rights to economic resources, as well as access to basic services, ownership and control over land and other forms of property, inheritance, natural resources, appropriate new technology and financial services, including microfinance	Increased access to basic services which includes clean cooking technology
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B.6.1. Explanation of methodological choices/approaches for estimating the SDG Impact

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SDG 13: Take urgent action to combat climate change and its impacts

Parameter: Emission reductions achieved by fuelwood savings at household level

As per TPDDTEC (V3.1) methodology, when the baseline fuel and the project fuel are the same and the baseline emission factor and project emission are considered the same, the overall GHG reductions achieved by the project activity in year y are calculated as follows:

$$ER_y = \sum_{b,p} (N_{p,y} * U_{p,y} * P_{p,b,y} * NCV_{b,fuel} * (f_{NRB,b,y} * EF_{fuel,CO2} + EF_{fuel,nonCO2})) - \sum LE_{p,y}$$

Where;

- $\sum_{b,p}$ Sum over all relevant (baseline b/project p) couples
- $N_{p,y}$ Cumulative number of project technology-days included in the project database for project scenario p against baseline scenario b in year y
- $U_{p,y}$ Cumulative usage rate for technologies in project scenario p in year y, based on cumulative adoption rate and drop off rate revealed by usage survey (fraction)
- $P_{p,b,y}$ Specific fuel savings for an individual technology of project p against an individual technology of baseline b in year y, in tons/day, as derived from the statistical analysis of the data collected from the field tests

$NCV_{b,fuel}$	Net calorific value of the fuel that is substituted or reduced (IPCC default for firewood, 0.015 TJ/ton)
$f_{NRB,b,y}$	Fraction of biomass used in year y for baseline scenario b that can be established as non-renewable biomass
$EF_{fuel,CO2}$	CO ₂ emission factor of the fuel that is substituted or reduced (112 tCO ₂ /TJ for Firewood/Firewood waste)
$EF_{fuel,nonCO2}$	Non-CO ₂ emission factor of the fuel that is reduced
$\Sigma LE_{p,y}$	Leakage for project scenario p in year y (tCO ₂ e/yr)

Number of project devices commissioned (N_y) will be monitored through database and monitoring surveys. The project-technology days ($N_{p,y}$) will be calculated by multiplying the number of project devices with number of operational days.

Annual usage surveys will be conducted to determine the usage rate of project devices ($U_{p,y}$). Sampling will be done for each age group of project population and weighted average will be calculated.

Quantity of fuel consumed in baseline scenario and project scenario will be determined by Kitchen Performance Tests (KPT). Specific fuel savings for a single stove ($P_{p,b,y}$) will be calculated based on the results as per the formula: $P_{p,b,y} = P_{b,y} - P_{p,y}$

Fraction of non-renewable biomass ($f_{NRB,b,y}$) is determined as using ex-ante option at the validation stage and will be fixed as 0.814 for the crediting period. The parameter has been calculated as per as per "Tool 30: calculation of the fraction of non-renewal biomass"; version 3.0.

For emission factors for fossil fuels displaced by the project stoves ($EF_{fuel,CO2}$ and $EF_{fuel,nonCO2}$) and net calorific value of the non-renewable woody biomass ($NCV_{b,fuel}$), default values will be applied; i.e. 112 tCO₂/TJ for CO₂, 14.2 tCO₂e/TJ (AR5 GWP) for non-CO₂ emissions.

Leakage ($\Sigma LE_{p,y}$)

The project proponent should investigate the following potential sources of leakage:

- a. The displaced baseline technologies are reused outside the project boundary in place of lower emitting technology or in a manner suggesting more usage than would have occurred in the absence of the project.

Reason for not considering: The project aims to displace traditional three stones stove that is already widely used in rural Zambia. It is unlikely that the project will cause the households outside the project boundaries to use higher emitting technologies or to increase their fuelwood consumption.

- b. Non-project users who previously used lower emitting energy sources use the non-renewable biomass or fossil fuels saved under the project activity.

Reason for not considering: Majority of the households rely on fuelwood in rural Zambia as per 2018 Demographic and Health Survey done by Central Statistics Office (Figure.1)¹¹. Only 13% of the population has been reported to have access to clean cooking services in the country 2018¹². No possible leakage is foreseen due to this case.

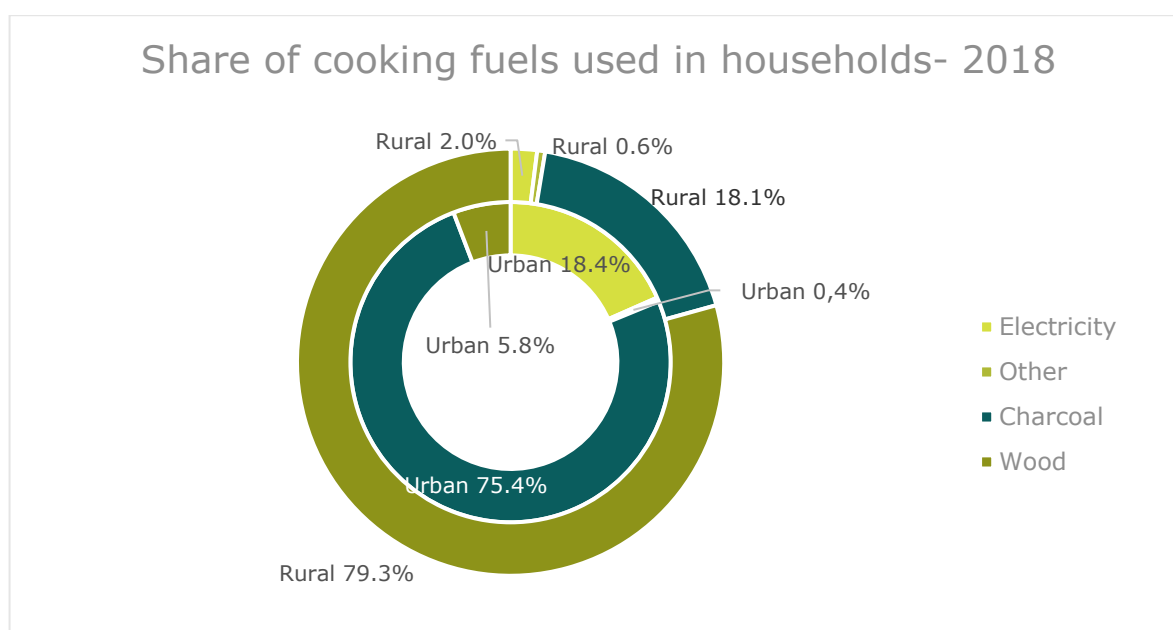


Figure 2. Share of cooking fuels used in households, Zambia, 2018.

¹¹ <https://dhsprogram.com/pubs/pdf/FR361/FR361.pdf>

¹² <https://trackingsdg7.esmap.org/country/zambia>

- c. The project significantly impacts the NRB fraction within an area where other CDM or VER project activities account for NRB fraction in their baseline scenario.

Reason for not considering: Zambia's forests cover around 61% of the country's land area and provides energy in the form of fuelwood and charcoal for about 80% of Zambians. Deforestation and forest degradation are the greatest contributors to GHG emissions¹³. The forest cover in Zambia has been constantly dropping since 1990 and 2.6 Mha of forest land has been lost in last 30 years. The rate of deforestation has accelerated between 2015-2020. 1.1 Mha of total forest lost has been lost with an average rate of 188kha annually (0.4% of the total forest area) in the same period (Global Forest Resources Assessment (FRA) 2020- Zambia Report¹⁴).

Since firewood is collected from forest by the majority of rural population, no significant impact is expected in NRB fraction in other areas.

- d. The project population compensates for loss of the space heating effect of inefficient technology by adopting some other form of heating or by retaining some use of inefficient technology.

Reason for not considering: Only 10% of the target population cook inside the house as per 2018 Demographic and Health Survey⁹(Figure.2). Considering the climate of the country (tropical or sub-tropical depending on the altitude), space heating is not required in most of the country.

¹³ <https://documents1.worldbank.org/curated/en/571651580133910005/pdf/Zambia-Country-Forest-Note-Towards-a-Sustainable-Way-of-Managing-Forest.pdf>

¹⁴ <http://www.fao.org/3/cb0092en/cb0092en.pdf>

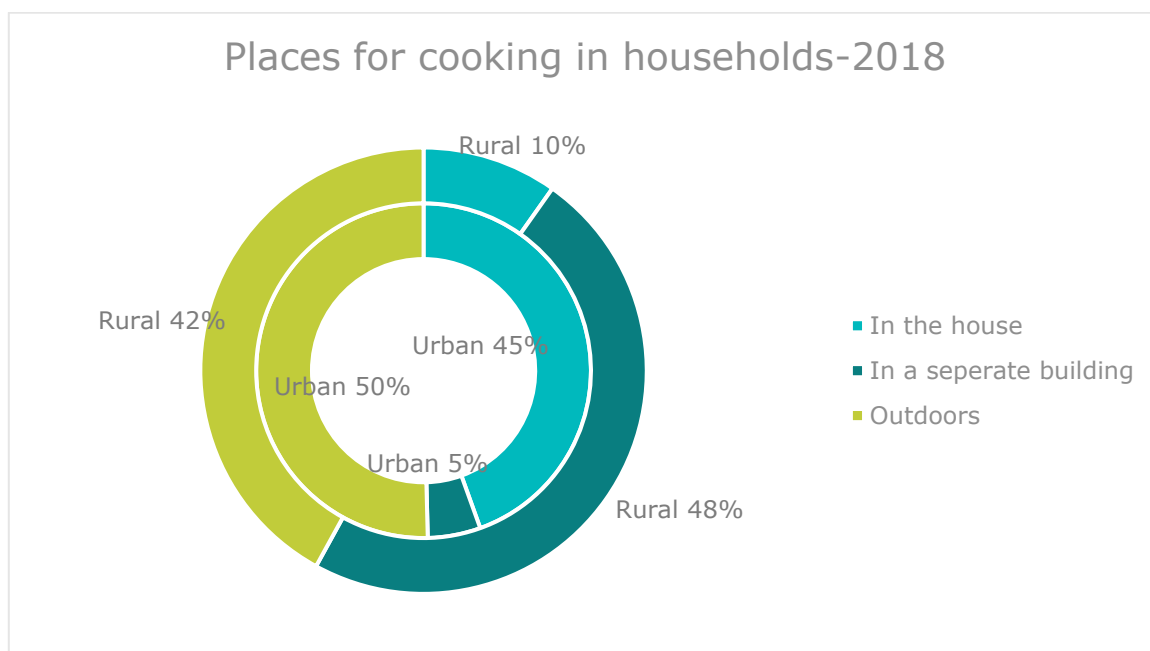


Figure 3. Places for cooking in households in Zambia, 2018

- e. By virtue of promotion and marketing of a new technology with high efficiency, the project stimulates substitution within households who commonly used a technology with relatively lower emissions, in cases where such a trend is not eligible as an evolving baseline.

Reason for not considering: This option is not considered as a potential source of leakage as the majority of targeted households use fuelwood and three stone fire for cooking.

SDG 15. Protect, restore, and promote sustainable use of terrestrial ecosystems, sustainably manage forests, combat desertification, and halt and reverse land degradation and halt biodiversity loss

Parameter: Amount of firewood saved by the project stoves per year.

Project database and monitoring survey will be used to calculate total quantity of firewood saved by the project compared to baseline scenario. The following formula will be used to calculate the total amount of firewood saved:

$$\text{Total fuel savings} = P_{p,b,y} * N_{p,y}$$

Where;

$N_{p,y}$ Project technology-days in the project database for project scenario p

through year y

$P_{p,b,y}$ Quantity of firewood that is saved in project scenario p compared to baseline scenario b during year y

$$P_{p,b,y} = P_{b,y} - P_{p,y}$$

$P_{b,y}$ Quantity of firewood that is consumed in baseline scenario b during year y

$P_{p,y}$ Quantity of firewood that is consumed in project scenario p during year y

SDG 3. Ensure healthy lives and promote well-being for all at all ages

Parameter: Proportion of households who perceive reduced indoor air pollution

The beneficiaries of project stoves will be asked to evaluate any improvement in their health conditions compared to baseline situation during the household survey. By reduction of smoke due to the usage of project stoves, respiratory diseases such as itchy eyes or coughing are expected to lessen.

SDG 5. Achieve gender equality and empower all women and girls

Parameter: Proportion of households who perceive reduced time for collecting wood and cooking

During the household survey, questions about the perception of time spent on cooking and fuelwood collection will be asked to women in order to understand whether using project stoves saves time. If the response is positive, the surveyors will further explore for what kind of activities people are using in their time saved.

Parameter: Number of women artisans who received training

Training records will be collected to monitor the number of women artisans who received training.

SDG 8. Promote inclusive and sustainable economic growth, employment and decent work for all

Parameter: Number of temporary and permanent jobs created

Number and type of jobs created will be recorded with employment status and duration.

SDG 1. No Poverty

Parameter: All men and women, in particular the poor and the vulnerable, have access to basic services and appropriate new technology.

Number of clean cooking technology devices implemented under the project.

B.6.2. Data and parameters fixed ex ante

SDG13

Data/parameter	$EF_{fuel, non-CO2}$
Unit	tCO ₂ /TJ
Description	Non-CO ₂ emission factor arising from use of fuels in baseline and project scenario
Source of data	Calculated by using 2006 IPCC Guidelines for National Greenhouse Gas Inventories, Table 2 and Table 3. ¹⁵ and WG1_AR5 Report, Chapter8 (Table 8.7).
Value(s) applied	CH ₄ : 0.455 tCH ₄ /TJ and N ₂ O: 0.006 tN ₂ O/TJ GWP _{CH₄} :28 GWP _{N₂O} : 265
Choice of data or Measurement methods and procedures	Default value as 14.2 tCO ₂ e/TJ
Purpose of data	Calculation of emission reductions
Additional comment	The following GWP (100 years) ¹⁶ are applied: 28 for CH ₄ and 265 for N ₂ O

Data/parameter	$NCV_{b, fuel}$
Unit	TJ/tonne

¹⁵ https://www.ipcc-nggip.iges.or.jp/public/gp/bgp/2_2_Non-CO2_Stationary_Combustion.pdf

¹⁶ <https://globalgoals.goldstandard.org/applicability-of-global-warming-potential-for-gold-standard-for-the-global-goals-projects/>

Description	Net calorific value of the non-renewable woody biomass that is substituted
Source of data	IPCC default value
Value(s) applied	0.0156
Choice of data or Measurement methods and procedures	Default value
Purpose of data	Calculation of emission reductions
Additional comment	N/A

B.6.3. Ex ante estimation of SDG Impact

>>

SDG 13: Take urgent action to combat climate change and its impacts

Parameter: Emission reductions achieved by fuelwood savings at household level

Ex-ante calculation for a project stove by using the parameter values fixed ex-ante and the preliminary values defined for monitored parameters in B.7.1 is demonstrated as follows:

Parameter	Unit	Data Source	Value
$N_{p,y}$	Number	Project technology-days in the project database for project scenario p through year y	Assumed as 365 days per year for each stove.
$U_{p,y}$	Fraction	Assumed. (Sample surveys for CDM PoA 8060)	0.904
$P_{p,b,y}$	t/hh/d	Assumed Field performance test results (KPT)	$P_{p,b,y} = P_{b,y} - P_{p,y}$ $P_{p,b,y} = 0.0148 \text{ t/hh/d} - 0.0043 \text{ t/hh/d} = 0.0105 \text{ t/hh/d}$
$NCV_{b,fue}$ I	TJ/t	Default value for firewood	0.0156

Parameter	Unit	Data Source	Value
$f_{NRB,b,y}$	Fraction	NRB Report	0.8140
EF_{fuel,CO_2}	tCO ₂ /TJ	Default value for firewood	112
$EF_{fuel,nonCO_2}$	tCO ₂ /TJ	Default value for firewood	CH ₄ =0.445*28=12.7 NO ₂ =0.006*265=1.5 ⇒ 14.2
$\Sigma LE_{p,y}$	tCO ₂	Assumed to be zero as per the methodology	0

$$ER_y = \Sigma b,p (N_{p,y} * U_{p,y} * P_{p,b,y} * NCV_{b,fuel} * (f_{NRB,b,y} * EF_{fuel,CO_2} + EF_{fuel,nonCO_2}) - \Sigma LE_{p,y})$$

Emission reductions achieved per project stove is calculated as follows:

$$ER_y = 365 * 0.904 * (0.0105 \text{ tonnes/hh/d}) * (0.0156 \text{ TJ/tonnes}) * ((0.8140 * 112) + 14.2) \text{ tCO}_2/\text{TJ} - 0 = 5.67 \text{ tCO}_2/\text{y/stove}$$

SDG 15. Protect, restore and promote sustainable use of terrestrial ecosystems, sustainably manage forests, combat desertification, and halt and reverse land degradation and halt biodiversity loss

Parameter: Amount of firewood saved by the project stoves per year

Amount of firewood saved will be calculated by the formula: $P_{p,b,y} = P_{b,y} - P_{p,y}$

P_{b,y}	Baseline fuel consumption (Tons/HH/Yr)	5.3838
P_{p,y}	Project fuel consumption (Tons/HH/Yr)	1.5658
P_{p,b,y}	Calculated (Tons/HH/Yr)	P_{p,b,y} = P_{b,y} - P_{p,y} 3.8180

For SDG 3 and SDG 5, the results of household survey will be used, no calculation is applicable.

For SDG 8 and SDG 1, the project database will be used. Additionally for SDG 8, training records will also be used.

B.6.4. Summary of ex ante estimates of each SDG outcome

Year	Baseline estimate	Project estimate	Net benefit
Year 2020	0	0	141,203
Year 2021	0	0	282,407
Year 2022	0	0	282,407
Year 2023	0	0	282,407
Year 2024	0	0	282,407
Total	0	0	1,270,831
Total number of crediting years			
Annual average over the crediting period	0	0	254,166

SDG 15 Parameter: Amount of firewood saved by the project stoves per year (tonnes/y)

Year	Baseline estimate	Project estimate	Net benefit
Year 2020	134593	39,145	95449
Year 2021	269187	78,290	190898
Year 2022	269187	78,290	190898
Year 2023	269187	78,290	190898
Year 2024	269187	78,290	190898
Total	1211341	352,304	859039
Total number of crediting years			
Annual average over the crediting period	242268	70461	171807

SDG 3 Parameter: Proportion of households who perceive reduced indoor air pollution

Year	Baseline estimate	Project estimate	Net benefit
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Year 2020	0	100%	100%
Year 2021	0	100%	100%
Year 2022	0	100%	100%
Year 2023	0	100%	100%
Year 2024	0	100%	100%
Total	0	100%	100%
Total number of crediting years			
Annual average over the crediting period	100%		

SDG 5 Parameter: Proportion of households who perceive reduced time for collecting wood and cooking

Year	Baseline estimate	Project estimate	Net benefit
Year 2020	0	100%	100%
Year 2021	0	100%	100%
Year 2022	0	100%	100%
Year 2023	0	100%	100%
Year 2024	0	100%	100%
Total	0	100%	100%
Total number of crediting years			
Annual average over the crediting period	100%		

SDG 5 Parameter: Number of women artisans who received training

Year	Baseline estimate	Project estimate	Net benefit
Year 2020	0	20	20

Year 2021	0	20	20
Year 2022	0	0	0
Year 2023	0	0	0
Year 2024	0	0	0
Total	0	40	40
Total number of crediting years			
Annual average over the crediting period	8		

SDG 8 Parameter: Number of temporary and permanent jobs created

Year	Baseline estimate	Project estimate	Net benefit
Year 2020	0	25	25
Year 2021	0	25	25
Year 2022	0	0	0
Year 2023	0	0	0
Year 2024	0	0	0
Total	0	50	50
Total number of crediting years			
Annual average over the crediting period	10		

SDG 1 Parameter: Number of households with basic services

Year	Baseline estimate	Project estimate	Net benefit
2022	0	25,000	25,000
2023	0	50,000	50,000
2024	0	50,000	50,000

2025	0	50,000	50,000
2026	0	50,000	50,000
Total	0	50,000	50,000
Total number of crediting years			
Annual average over the crediting period	10,000		

B.7. Monitoring plan

B.7.1. Data and parameters to be monitored

SDG 13

Data / Parameter	$N_{p,y}$
Unit	Project technologies credited (unit)
Description	Technologies in the project database for project scenario p through year y
Source of data	Project Database Records
Value(s) applied	50,000 (assumed)
Measurement methods and procedures	Detailed sales information will be collected for each customer, either using electronic or paper-based means. On electronic means, the data will be collected by SMS (i.e. mobile phone 'short message service' text) or Information and Communication Technologies ('ICT' – such as PDAs). Information that is entered into the database includes the name of the customer, address/ description of location, contact telephone number(s), unique serial number of the stove, retailer ID, and date of purchase. Sales information submitted via SMS/ ICT will automatically enter the database. Written registration cards will be entered manually into the same database.

Monitoring frequency	Continuously
QA/QC procedures	Project distribution staff will spot-check end-users to verify that information submitted was factual.
Purpose of data	Cross checking Calculation of emission reductions
Additional comment	N/A

SDG 13

Data / Parameter	$U_{p,y}$
Unit	Percentage
Description	Cumulative usage rate for technologies in project scenario p during year y, based on cumulative installation rate and drop-off rate.
Source of data	Usage Survey
Value(s) applied	90.00% (assumed)
Measurement methods and procedures	The proportion of sampled ICS found to be in operation during each monitoring period will be applied to the total number of stoves accounted for emission reductions. Minimum 100 samples will be surveyed. Separate samples of at least 30 shall be taken for each age group of project population and weighted average factor for overall project population shall be used.
Monitoring frequency	Annually
QA/QC procedures	The unique reference number of each stove shall be logged in the monitoring database showing the total number of stoves. Data from the sampling plan will be collected in each monitoring period by trained project staff and applied in the emissions reduction calculations. Internal cross-checks by the project developer or project implementer will be undertaken as QC
Purpose of data	Calculation of emission reductions

Additional comment	N/A
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Data / Parameter	fNRB
Unit	Fraction
Description	Fraction of woody biomass saved by the project activity in year y that can be established as non-renewable biomass
Source of data	Calculated
Value(s) applied	0.814
Measurement methods and procedures	Calculated as per "Tool 30: calculation of the fraction of non-renewal biomass"; version 3.0
Monitoring frequency	Fixed by baseline study for a given crediting period
QA/QC procedures	Transparent data analysis and reporting
Purpose of data	Calculation of baseline emissions
Additional comment	N/A

Data / Parameter	$P_{b,y}$
Unit	t/hh/day
Description	Quantity of firewood that is consumed in baseline scenario b during year y
Source of data	Baseline FT, Baseline FT updates and any applicable adjustment factors
Value(s) applied	0.0148 t/hh/day
Measurement methods and procedures	Determined by the baseline KPT carried out sampling the households without the project stove from targeted population. For independent sampling, the sample size of stove users shall be chosen for a 90/30 precision (90% confidence interval+ and 30% margin of error). In cases

	where the result indicates that 90/30 precision is not achieved, the lower limit of 30% confidence interval of the parameter value will be chosen as an alternative to repeating the survey efforts to achieve the 90/30 precision.
Monitoring frequency	Whenever required to update the baseline.
QA/QC procedures	Transparent data analysis and reporting, the latest protocol for Kitchen Performance Test ¹⁷ will be applied.
Purpose of data	Calculation of emission reductions
Additional comment	The baseline is fixed ex-ante during the crediting period as the project is targeting non-industrial application (TPDDTEC, Page 8). A single baseline fuel consumption parameter is weighted to be representative of baseline technologies being compared for project crediting.

Data / Parameter	$P_{p,y}$
Unit	t/hh/day
Description	Quantity of firewood that is consumed in project scenario p during year y
Source of data	Project FT, and any applicable adjustment factor
Value(s) applied	0.0043 t/hh/day
Measurement methods and procedures	Ex ante: Calculated based on 34.5% thermal efficiency reported Ex Post: The latest protocol for Kitchen Performance Test¹⁸ will be applied. The sample size of stove users shall be chosen for a 90/30 precision (90% confidence interval+ and 30% margin of error). In cases where the

¹⁷ <https://www.cleancookingalliance.org/binary-data/DOCUMENT/file/000/000/604-1.pdf>

¹⁸ <https://www.cleancookingalliance.org/binary-data/DOCUMENT/file/000/000/604-1.pdf>

	result indicates that 90/30 precision is not achieved, the lower limit of 30% confidence interval of the parameter value will be chosen as an alternative to repeating the survey efforts to achieve the 90/30 precision.
Monitoring frequency	At least once every two years (biennial)
QA/QC procedures	Transparent data analysis and reporting
Purpose of data	Calculation of emission reductions
Additional comment	A single project fuel consumption parameter is weighted to be representative of the quantity of project technologies of each age being credited in a given project scenario

Data / Parameter	$LE_{p,y}$
Unit	tCO ₂ e
Description	Leakage in project scenario p during year y
Source of data	Baseline and monitoring surveys
Value(s) applied	0
Measurement methods and procedures	Measured through baseline surveys and monitoring surveys
Monitoring frequency	Biennially
QA/QC procedures	Transparent data analysis and reporting
Purpose of data	Calculation of leakage emissions
Additional comment	

Data / Parameter	η_{new}
Unit	Percent
Description	Efficiency of the ICS installed as a part of the project activity.
Source of data	Water Boiling Tests following the latest WBT protocol
Value(s) applied	34.5-

Measurement methods and procedures	As per Annex 8 of TPDDTEC(V3.1): • The minimum sample size of each age group shall comply with the 90/10 rule. • The tests will be carried out in the last 3 months of the monitoring period. • Samples will be selected as per the Guidelines for sampling and surveys for CDM project activities and programme of activities.
Monitoring frequency	Annually
QA/QC procedures	Transparent data analysis and reporting
Purpose of data	Calculation of project fuel consumption
Additional comment	Only applicable when the ageing test approach (Annex 8 of TPDDTEC) is preferred.

Data / Parameter	μ_y
Unit	Fraction
Description	Adjustment factor continued use of pre-project devices during the year y
Source of data	Monitoring surveys
Value(s) applied	-
Measurement methods and procedures	Any of the following approach with justification on how the chosen option fits within project circumstances • Measurement campaigns shall be undertaken using data loggers such as stove utilization monitors (SUMs) which can log the operation of all devices in order to determine the average device utilization intensity • Monitoring surveys to capture cooking habits and stove usage of households in the region, including quantification of use of baseline devices, by formulating questions and/or collecting evidences to determine the

	frequency of usage of both the project devices and baseline devices. • Monitoring surveys to capture the number of meals cooked
Monitoring frequency	Annual
QA/QC procedures	Transparent data analysis and reporting
Purpose of data	Calculation of emission reductions
Additional comment	Only applicable when the ageing test approach (Annex 8 of TPDDTEC) is preferred.

SDG 3

Data / Parameter	Proportion of households who perceive reduced indoor air pollution
Unit	Fraction
Description	Confirmation of reduced indoor air pollution due to the use of project stove by the households
Source of data	Household Surveys
Value(s) applied	100% (assumed)
Measurement methods and procedures	Questions about perceived indoor air pollution during cooking with the projects stoves will be asked to the households. The proportion of households who confirms that the project stoves reduce air pollution will be accounted.
Monitoring frequency	Annually/biennially
QA/QC procedures	N/A
Purpose of data	Assessment of contribution to SDG 3
Additional comment	N/A

SDG 5

Data / Parameter	Proportion of households who perceive reduced time for collecting wood and cooking
Unit	Fraction
Description	Confirmation of time saving due to the use of project stove by the households
Source of data	Household Surveys
Value(s) applied	100% (assumed)
Measurement methods and procedures	Questions about the change in time spent on domestic works such as cooking and fuelwood collection will be asked to the households. The proportion of households who confirms that the project stoves saved time will be accounted.
Monitoring frequency	Annually/biennially
QA/QC procedures	N/A
Purpose of data	Assessment of contribution to SDG 5
Additional comment	N/A

Data / Parameter	Number of woman artisans who received training
Unit	Number
Description	Count of woman artisans attended training sessions for building the project stoves from metal stove kit.
Source of data	Training records
Value(s) applied	To be reported in monitoring report
Measurement methods and procedures	Number of woman artisans who have attended the trainings for building the project stoves will be recorded.
Monitoring frequency	Annually/biennially
QA/QC procedures	N/A
Purpose of data	Assessment of contribution to SDG 5
Additional comment	N/A

SDG 8

Data / Parameter	Number of temporary and permanent jobs created
Unit	Number
Description	Temporary and permanent jobs created for building stove, maintenance, repair, sale and distribution.
Source of data	Employment records and contracts
Value(s) applied	To be reported in monitoring report
Measurement methods and procedures	The employment records including sex of the worker and contracts will used to monitor the parameter.
Monitoring frequency	Annually/biennially
QA/QC procedures	N/A
Purpose of data	Assessment of contribution to SDG 8
Additional comment	N/A

SDG 1

Data / Parameter	Number of households with basic services
Unit	Number
Description	Number of households which did not have clean cooking technology in pre-project scenario, have now access to clean cookstoves
Source of data	Project database
Value(s) applied	To be reported in monitoring report
Measurement methods and procedures	Registration records to be cross verified with number of operational stoves
Monitoring frequency	Annually/biennially
QA/QC procedures	N/A
Purpose of data	Assessment of contribution to SDG 1
Additional comment	N/A

B.7.2. Sampling plan

>>The sampling frame is the project boundaries, including the houses where the stoves have been installed. The target population are the households in rural Zambia. The end users who received the Project stoves will be recorded to the sales database. To ensure a random selection of ICS, random number generators shall be applied. Each ICS in the target population is uniquely identifiable by its unique ID number. Each ICS can thus be allocated a Sample Selection Number in each monitoring period, starting at 1 and increasing up to the total number of ICS in the Database for that pre-defined sampling frame. Applying the random number generators, the ICS can then be randomly chosen from the defined population up to the required sample size.

Baseline Survey:

For baseline survey, samples will be randomly selected among households without the project technology. The minimum sample size required by the methodology is 100 for population over 1000.

Kitchen Performance Test:

The sampling method for $P_{b,y}$ and $P_{p,y}$ is independent Sampling which is households sampled in the project situation are different from households sampled in the baseline situation.. Sample size must be sufficient to ensure the confidence/precision level of 90/30 for quantity of fuelwood consumed in baseline and project scenario. The minimum sample size recommended by the methodology is 30 for KPTs.

The baseline KPT was performed with 35 samples that were randomly selected from 110 end users in baseline survey. The KPT was performed on houses with similar socioeconomic characteristics as the target population. After baseline KPT was finalized, the households have been provided with the project stoves for the purpose of the test. The project KPT will be run after allowing sufficient time to the subjects to get used to the project stoves.

Subsequent sampling for updating $P_{p,y}$ will be done by stratified sampling method per stove ages. The registered stoves will be grouped by age from the project database. The sample size for conducting KPT will comply with 90/30 confidence/precision required by the methodology.

Water Boiling Test:

PP may opt to follow aging test approach for project fuel updates as per Annex.8 of the methodology. In that case, annual WBTs will be conducted with a representative sample of each age group instead of biennial KPTs. The first WBT will be carried out along with the project KPT prior to the first issuance in the field or laboratory, following the latest version of WBT protocol by an independent expert or entity. Subsequent WBTs will be carried out for monitoring the degradation of efficiency of ICS on annual basis. This shall be done during the last 3 months of monitoring provided the chosen period is representative of annual conditions.

Sample selection shall follow the recent Guidelines for Sampling and Surveys for CDM Project Activities and Programme Activities. The sample size shall comply with 90/10 confidence/precision level for each age group.

Usage Rate:

For $U_{p,y}$, simple random sampling method will be followed. Sample size for usage rate is minimum 100 for group sizes larger than 1,000.

Usage Rate Monitoring guidelines identifies the requirements as follows:

Level	Applicability	Claimable usage rate	Requirements
Mandatory	Mandatory	Upto maximum 75%	Define use and non-use

			• In person household usage survey • Verification of accuracy of results
Good Practice	Optional	Upto maximum 90%	• Field team training and supervision • End -user training and follow ups • Awareness campaign
Best Practice	Optional	Above 90%	• Continuous use monitoring

Monitoring Surveys:

The monitoring survey investigates changes over time in the project scenario by surveying end-users with project technology on an annual basis. Annual trends in end user characteristics such as stove use, fuel consumption and seasonal variations are predicted during the survey. Whether the baseline technology is still in use will be checked and recorded as well.

Leakage shall be assessed biennially as per the methodology. It will be done along with the monitoring survey

B.7.3. Other elements of monitoring plan

>>

The Monitoring Plan applied involves a number of key elements that ensure high-quality, unbiased and reliable information regarding the performance of the project in terms of implementation and outcomes, and for the purposes of calculating Verified Emission Reductions (VERs) on the basis of the amount of non-renewable biomass saved by the ICS in the project activity. The key elements are the following:

- Data collection procedures
- Distribution and Monitoring Database
- Spot Checking of ICS (ongoing)
- Sample Plan for the Monitoring Survey
- Data Quality, Consistency and Duplication Checks
- Monitoring Reporting

Below is the description of the steps in monitoring plan.

1. **Registration of stove:** Project Implementer will collect/receive the necessary information requested in the Registration process from the user. Means of collecting this information may be through a hard copy form or ICTs by project implementer's staff, retailers, end-users or partner organization's staff, or through the use of ICTs or SMS. Project Implementers' staff shall double check the accuracy of information provided, and request for field staff additional clarifications if needed. Following information will be recorded:

- Serial Number
- Contact details of the user
- Date of installation
- GPS Coordinates
- Number of stoves installed

Every beneficiary of an ICS will sign a user agreement (paper and/or electronic version) with Project Developer. A unique serial number will be allocated to each stove and the number is indicated on the user agreement.

2. **Data logged into database:** Project Implementer's trained staff will input the data in the database either manually (if data collected using hard copy form) or this will be automatically input if data was collected using ICTs or SMS. Project Implementer staff shall double check the information included on the database and check for duplications. Any duplicate information shall be investigated and errors corrected or excluded from the database if it is a true duplicate entry.
3. **Spot- checking (continuously):** Project Implementer field staff will randomly select households included in the database and visit them to cross-check the information on the database with the factual evidence in the field. Any inconsistencies found (eg. change in the address of a user) will be updated on the database, and in the case ICS are found to be no longer in use, they will be clearly marked as such and excluded from emission reductions calculations.
4. **Monitoring:** Project Implementer will follow the requirements as per methodology requirements to collect the necessary information for a monitoring report.

5. **Preparation of monitoring report:** Project Implementers or Project Developer will prepare the final monitoring report to be provided to the verifier VVB for verification of emission reductions.

Project Developer will coordinate and manage Project Implementer and assist them in implementing each element of the monitoring plan.

Biennial Project FT Update:

The PFT update is an extension of the project KPT and done biennially. The test provides a fuel consumption assessment representative of project technologies currently in use and predicts any changes in the project scenario over time as project stove age and new customers are added.

Annual Monitoring and Usage Surveys:

According to the methodology, end users can be surveyed at any time(s) throughout the year with care taken to collect information pertaining to seasonal variations in stove and fuel use patterns. Monitoring surveys can be conducted with usage survey participants that are currently using the project stove. Visual inspection of the premises to see if the project stoves are operational and in use and interview with end user will be done during the survey.

In case WBTs opted:

If aging test approach is followed, Project proponent will have to monitor parallel use of baseline stove in the project scenario by any of the following options:

- Measurement campaigns shall be undertaken using data loggers such as stove utilization monitors (SUMs) which can log the operation of all devices in order to determine the average device utilization intensity
- Monitoring surveys to capture cooking habits and stove usage of households in the region, including quantification of use of baseline devices, by formulating questions and/or collecting evidences to determine the frequency of usage of both the project devices and baseline devices.
- Monitoring surveys to capture the number of meals cooked

Baseline FT Update:

If baseline is required to be updated any time during the crediting period, baseline performance field tests will be carried out as per described above

SECTION C. DURATION AND CREDITING PERIOD

C.1. Duration of project

C.1.1. Start date of project

>>

The installation of first stove under the project is 11/12/2020

C.1.2. Expected operational lifetime of project

>>15 years

C.2. Crediting period of project

C.2.1. Start date of crediting period

>>

16/10/2021

The stoves included under the project activity are being transferred from VCS project activity "Installation of High Efficiency Wood Burning Cookstoves in Zambia - Project 2" (VCS reference number 2371) for which credits have been issued until 15th October 2021.

C.2.2. Total length of crediting period

>>

16/10/2021 to 08/12/2035

5 years renewable twice, 15 years of total crediting period. As approximately one year (309 days) has already been claimed under VCS, the total crediting period under GS4GG will be 14 years.

SECTION D. SUMMARY OF SAFEGUARDING PRINCIPLES AND GENDER SENSITIVE ASSESSMENT

D.1. Safeguarding Principles that will be monitored

A completed Safeguarding Principles Assessment is in [Appendix 1](#), ongoing monitoring is summarised below.

Principles Mitigation Measures added to the Monitoring Plan

Necessary precautions shall be undertaken by the Project Implementers

Principle 3 during the stove installation. First aid kits shall be in available in workshops. Trainings will be provided for all workers

Principle 9.7 The users will be trained on how to harvest trees and which parts could be used as fuelwood.

D.2. Assessment that project complies with GS4GG Gender Sensitive requirements

Question 1 - Explain how the project reflects the key issues and requirements of Gender Sensitive design and implementation as outlined in the Gender Policy?	The project integrates gender sensitive design and gender aspect in all phases of the project. Women and youth will benefit mostly from the efficient cookstoves installed in households and therefore their feedback will be considered during the design and implementation of the project. Moreover, the project plans to organize trainings for women and community artisans on making bricks and building their own stoves. This will increase the participation of women and empower them in the society.	
Question 2 - Explain how the project aligns with existing country policies, strategies and best practices	The first National Gender Policy for Zambia was adopted in March 2000. The overall goal of the policy was to facilitate effective and equitable participation of both men and women in the social economic development of the country. This was followed by the establishment of institutional framework that led to the creation of Ministry of Gender and Child Development in March 2012. National Gender Policy was revised on 2014 to address the gaps identified in the	

process of implementation and emerging gender related issues (National Gender Policy 2014¹⁹).

According to the 2010 Census of Population and Housing Report, females account 51% of the Zambia's human population. The same report shows that 61% of the population lives in rural areas. The discrimination against women in the country is embodied in traditional rules and practices. Women are generally restricted from participating in important decision at community and household levels. Their level of education is a constrain to work in high paying sectors or technical jobs.

Traditionally, women and children especially girls collect firewood that is the major source of energy in most households especially in rural Zambia. They have to walk long distances as the source of wood fuel is usually far from settlements due to deforestation and other reasons. This makes children too tired to concentrate at school and increases women and girl's exposure to gender based violence as well as health problems such as persistent back aches, ill health among pregnant women and

¹⁹ <http://extwprlegs1.fao.org/docs/pdf/zam152916.pdf>

	snake bites. Furthermore, they are exposed to open flames and smoke during cooking which result in respiratory problems. The project addresses gender issues related with energy by installing improved cookstove technologies for households. Women and girls, being the main beneficiaries of the project are actively involved in all phases of the project including the design of cookstoves and participating stakeholder consultation. The project will further provide training to women artisans to make bricks and build their own stoves and will record the number of women employed. This empowers women by generating income and raise their status in the family.	
Question 3 - Is an Expert required for the Gender Safeguarding Principles & Requirements?	Gender issues raised by Gold Standard Safeguarding principles are all addressed under the Safeguarding Principles Assessment in Appendix.1. No risks have been foreseen by the stakeholders and project developer.	
Question 4 - Is an Expert required to assist with Gender issues at the Stakeholder Consultation?	As per Stakeholder Consultation Requirements, the project developer invites all stakeholders to give feedback on the design and the implementation of the project. During the consultation process, particular importance is given to the equal participation of women. A	

group of beneficiaries who have been using the technology have been invited to the live meeting and their trip was arranged by the project developer. Moreover, gender related organizations who actively work in rural areas have been invited to the meeting.

SECTION E. SUMMARY OF LOCAL STAKEHOLDER CONSULTATION

The below is a summary of the 2 step GS4GG Consultation for monitoring purposes. Please refer to the separate Stakeholder Consultation Report for a complete report on the initial consultation and stakeholder feedback round.

E.1. Summary of stakeholder mitigation measures

>>

During the stakeholder consultation, that the users of the project stove shall be trained for sustainable harvesting of trees. This has been included in the monitoring of Principle 9.7 above.

E.2. Final continuous input / grievance mechanism

Method	Include all details of Chosen Method (s) so that they may be understood and, where relevant, used by readers.
Continuous Input / Grievance Expression Process Book (mandatory)	The process book will be placed in the local office of C Quest Capital in Lusaka. Copies will also be available at local partners' offices.
GS Contact (mandatory)	help@goldstandard.org
Other	Tel (Zambia): 0977.636649 Tel (International): +1 202 416 2400 Cqc-operations@cqceuestcapital.com bmoser@cquestcapital.com

APPENDIX 1 - SAFEGUARDING PRINCIPLES ASSESSMENT

Complete the Assessment below and copy all Mitigation Measures for each Principle into [SECTION D](#) above. Please refer to the instructions in the [Guide to Completing](#) this Form below.

Assessment Questions/ Requirements	Justification of Relevance (Yes/potentially/no)	How Project will achieve Requirements through design, management or risk mitigation.	Mitigation Measures added to the Monitoring Plan (if required)
Principle 1. Human Rights			
1. The Project Developer and the Project shall respect internationally proclaimed human rights and shall not be complicit in violence or human rights abuses of any kind as defined in the Universal Declaration of Human Rights 2. The Project shall not discriminate with regards to participation and inclusion		1. The Republic of Zambia is a member of the United Nations and the African Union. It has ratified many UN Human Rights Conventions and thus has made binding international commitments to adhere to the standards laid down in these universal	NA

		human rights documents.²⁰ The project will be implemented under the national laws and will not lead to violations of human rights in any kind. 2. There is no limitation to the participation to the project.	
Principle 2. Gender Equality			
1. The Project shall not directly or indirectly lead to/contribute to adverse impacts on gender		National Energy Policy Report 2019 ²¹ states that most of the women in rural communities devote most of their	NA

²⁰ <http://www.claiminghumanrights.org/zambia.html>

²¹ https://www.moe.gov.zm/?wpfb_dl=51

equality and/or the situation of women 2. Projects shall apply the principles of nondiscrimination, equal treatment, and equal pay for equal work 3. The Project shall refer to the country's national gender strategy or equivalent national commitment to aid in assessing gender risks 4. (where required) Summary of opinions and recommendations of an Expert Stakeholder(s)		productive time collecting firewood to prepare a meal for the family. The project will improve the living conditions for women by the following: 1. The project aims to decrease the burden on women in the most vulnerable communities by reducing fuel wood consumption. The time spending for fuel wood collection and cooking will reduce. The women will have more time for other activities. The risk of being exposed to gender-based violence will decrease as well. 2. Both women and men will have equal access	
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		to the project stoves and other benefits. The project will provide job opportunities and aims to provide trainings to women artisans for building project stoves. Principles of equal treatment, equal pay for equal work will be strictly followed. 3. The Project respects the country's gender policy. The project addresses gender issues related with energy by installing improved cook stove technologies for households. Women and girls, being the main beneficiaries of	
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		the project are actively involved in all phases of the project including the design of cookstoves and participating stakeholder consultation. 4. N/A	
Principle 3. Community Health, Safety and Working Conditions			
1. The Project shall avoid community exposure to increased health risks and shall not adversely affect the health of the workers and the community		The project will follow the requirements in Occupational Health and Safety Act 2010²². The Project Implementers will provide safe workplace, machinery and equipment for the artisans.	Necessary precautions shall be undertaken by the Project Implementers during the stove installation. First aid kits shall be in available in workshops. Trainings will be provided for all workers.
Principle 4.1 Sites of Cultural and Historical Heritage			

²² <http://www.parliament.gov.zm/node/3409#:~:text=An%20Act%20to%20establish%20the,and%20suppliers%20of%20articles%2C%20devices%2C>

Does the Project Area include sites, structures, or objects with historical, cultural, artistic, traditional or religious values or intangible forms of culture?	No	The project does not involve and is not complicit in the alteration, damage or removal of any critical cultural heritage.	NA
>>			
Principle 4.2 Forced Eviction and Displacement			
Does the Project require or cause the physical or economic relocation of peoples (temporary or permanent, full or partial)?	No	The project does not require or cause the involuntary relocation of people. The project stoves will be installed in houses.	NA
>>			
Principle 4.3 Land Tenure and Other Rights			
a.Does the Project require any change, or have any uncertainties related to land tenure arrangements and/or access rights, usage rights or land ownership? b. For Projects involving land use tenure, are there any uncertainties with regards to land tenure, access rights, usage rights or land ownership?	No	The project does not require any change to land tenure arrangements and/or other rights such as resource access rights, community-based property rights and customary rights.	NA

>>			
Principle 4.4 - Indigenous people			
Are indigenous peoples present in or within the area of influence of the Project and/or is the Project located on land/territory claimed by indigenous peoples?	No	The project does not require any land for implementation.	NA
>>			
Principle 5. Corruption			
1. The Project shall not involve, be complicit in or inadvertently contribute to or reinforce corruption or corrupt Projects		Anti-corruption Act 2012 ²³ of Zambia provides domestication of the United Nations Convention Against Corruption, the African Union Convention on Preventing and Combating Corruption, the Southern African Development Community Protocol Against Corruption and other regional and	NA

²³ <http://www.parliament.gov.zm/node/3174>

		international instruments on corruption to which the country is a party. The project participants will not be involved, complicit or contribute towards corruption	
Principle 6.1 Labour Rights			
1. The Project Developer shall ensure that all employment is in compliance with national labour occupational health and safety laws and with the principles and standards embodied in the ILO fundamental conventions 2. Workers shall be able to establish and join labour organisations 3. Working agreements with all individual		1. Zambia ratified ILO N°87 Freedom of Association and Protection of the Right to organize convention in 1996²⁴. The project participants will employ all workers in accordance with all applicable national laws. 2. Zambia ratified ILO N°98 Right to organise and collective bargaining convention in 1996. The	

²⁴ https://www.ilo.org/dyn/normlex/en/f?p=1000:11200:0::NO:11200:P11200_COUNTRY_ID:103264

workers shall be documented and implemented and include: a) Working hours (must not exceed 48 hours per week on a regular basis), AND b) Duties and tasks, AND c) Remuneration (must include provision for payment of overtime), AND d) Modalities on health insurance, AND e) Modalities on termination of the contract with provision for voluntary resignation by employee, AND f) Provision for annual leave of not less than 10 days per year, not including sick and casual leave. 4. No child labour is allowed (Exceptions for		project participants will not restrict any workers from establishing and joining labour organisations. 3. Zambia ratified ILO N°29 Forced Labour Convention in 1996. All permanent workers will be provided with individual work agreements, including working hours, description of duties and tasks, remuneration, health insurance, termination of the contract, annual leave. 4. Zambia ratified ILO N°182 Worst Form of Child Labour Convention in 2001 and ILO No138 Minimum Age Convention in 1976. The project participants do not	
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children working on their families' property requires an Expert Stakeholder opinion) 5. The Project Developer shall ensure the use of appropriate equipment, training of workers, documentation and reporting of accidents and incidents, and emergency preparedness and response measures		engage in any form of child labour. 5. The project participants will provide safe working environment, machinery and appropriate equipment. Training will be carried out for the production of stoves and health and safety measures. Accidents and incidents will be monitored and reported.	
Principle 6.2 Negative Economic Consequences			
1. Does the project cause negative economic consequences during and after project implementation?	No	No risks are foreseen in terms of negative consequences for local economy. The project will be financially sustainable through the sale of cookstoves and carbon finance. The project will also create jobs for local people. Efficient biomass cookstoves are further promoted as a	
>>			

		policy measure as stated in National Energy Policy Report 2019 ²¹ under "Objective 4: To promote sustainable exploitation of biomass and alternative energy to wood fuel resources in order to increase socio-economic development." (page 33).	
Principle 7.1 Emissions			
Will the Project increase greenhouse gas emissions over the Baseline Scenario?	No	The project will not lead to any increase in greenhouse gas emissions. The project stoves will rather reduce emissions due to the increased thermal efficiency compared to the baseline stoves.	NA
>>		National Energy Policy Report 2019 ²¹ , wood fuel is stated to be the most widely used fuel for cooking. Its utilization is considered unsustainable because the harvest exceeds re-growth of biomass	

		contributing to climate change and negative health effects.	
Principle 7.2 Energy Supply			
Will the Project use energy from a local grid or power supply (i.e., not connected to a national or regional grid) or fuel resource (such as wood, biomass) that provides for other local users?	No	The project does not utilize any form of energy supply that is also being used by other users.	NA
>>			
Principle 8.1 Impact on Natural Water Patterns/Flows			
Will the Project affect the natural or pre-existing pattern of watercourses, ground-water and/or the watershed(s) such as high seasonal flow variability, flooding potential, lack of aquatic connectivity or water scarcity?	No	The project does not use any water.	NA
>>			
Principle 8.2 Erosion and/or Water Body Instability			
a. Could the Project directly or indirectly cause additional erosion and/or water body	No	a. The project includes installation of efficient cookstoves in households.	NA

instability or disrupt the natural pattern of erosion? b. Is the Project's area of influence susceptible to excessive erosion and/or water body instability?		No damage is foreseen for nature of soil or water bodies. b. The project stoves will be installed in end user premises. The area of influence thus is individual stove locations.	
>>			
Principle 9.1 Landscape Modification and Soil			
Does the Project involve the use of land and soil for production of crops or other products?	No	The project includes installation of efficient cookstoves in households. No use of land or soil for production of crops is applicable.	NA
>>			
Principle 9.2 Vulnerability to Natural Disaster			
Will the Project be susceptible to or lead to increased vulnerability to wind, earthquakes, subsidence, landslides, erosion, flooding, drought or other extreme climatic conditions?	No	The project includes installation of efficient cookstoves in households. No such risk is foreseen.	NA
>>			
Principle 9.3 Genetic Resources			

Could the Project be negatively impacted by or involve genetically modified organisms or GMOs (e.g., contamination, collection and/or harvesting, commercial development, or take place in facilities or farms that include GMOs in their processes and production)?	No	Not relevant	NA
>>			
Principle 9.4 Release of pollutants			
Could the Project potentially result in the release of pollutants to the environment?	No	The project includes installation of efficient cookstoves in households. Indoor air pollution in houses will be reduced compared to the baseline stoves. National Energy Policy Report 2019 ²¹ states that respiratory illnesses as a result of exposure to smoke from traditional cookstoves is mentioned as a negative impact to be mitigated (Section 2.9)	NA
>>			

Principle 9.5 Hazardous and Non-hazardous Waste			
Will the Project involve the manufacture, trade, release, and/ or use of hazardous and non-hazardous chemicals and/or materials?		Not relevant	
>>			
Principle 9.6 Pesticides & Fertilisers			
Will the Project involve the application of pesticides and/or fertilisers?		Not relevant	
>>			
Principle 9.7 Harvesting of Forests			
Will the Project involve the harvesting of forests?		The project aims to reduce firewood consumption by installing highly efficient cookstoves. This will reduce the harvest rate of forests. National Energy Policy Report 2019 ²¹ states that wood fuel is the most widely used fuel for cooking and its utilization is considered unsustainable because the harvest exceeds re-growth of biomass	
>>			

		contributing to climate change and negative health effects.	
Principle 9.8 Food			
Does the Project modify the quantity or nutritional quality of food available such as through crop regime alteration or export or economic incentives?		Not relevant	
>>			
Principle 9.9 Animal husbandry			
Will the Project involve animal husbandry?		Not relevant	
>>			
Principle 9.10 High Conservation Value Areas and Critical Habitats			
Does the Project physically affect or alter largely intact or High Conservation Value (HCV) ecosystems, critical habitats, landscapes, key biodiversity areas or sites identified?		Implementation of the project will have a positive impact on forests; thus, to conservation of HCV ecosystems, critical habitats, landscapes, key biodiversity areas or other sites, by reducing the firewood harvesting.	
>>			

		National Energy Policy Report 2019 ²¹ states that wood fuel is the most widely used fuel for cooking and its utilization is considered unsustainable because the harvest exceeds re-growth of biomass contributing to climate change and negative health effects.	
Principle 9.11 Endangered Species			
a. Are there any endangered species identified as potentially being present within the Project boundary (including those that may route through the area)? b. Does the Project potentially impact other areas where endangered species may be present through transboundary affects?		a. Not relevant as the project activity involves installation of energy efficient cookstoves b. No	NA
>>			

APPENDIX 2- CONTACT INFORMATION OF VPA IMPLEMENTER

Organization name	C-Quest Capital Stoves Asia Limited
Registration number with relevant authority	
Street/P.O. Box	
Building	Brumby Centre, Lot 42, Jalan Muhibbah
City	Labuan F.T.
State/Region	8700
Postcode	Malaysia
Country	Malaysia
Telephone	+6 087 593828
E-mail	cqc-operations@cquestcapital.com
Website	https://cquestcapital.com
Contact person	Ken Newcombe
Title	Director
Salutation	Dr.
Last name	Newcombe
Middle name	
First name	Ken
Department	
Mobile	+1-240-491-2650
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Personal e-mail	cqc-operations@cquestcapital.com

APPENDIX 3-SUMMARY OF APPROVED DESIGN CHANGES

Please refer to [Design Changes Requirements](#) for more information on procedures governing Design Changes

Revision History

Version	Date	Remarks
1.1	7 October 2020	Hyperlinked section summary to enable quick access to key sections Improved clarity on Key Project Information Inclusion criteria table added Gender sensitive requirements added Prior consideration (1 yr rule) and Ongoing Financial Need added Safeguard Principles Assessment as annex and a new section to include applicable safeguards for clarity Improved Clarity on SDG contribution/SDG Impact term used throughout Clarity on Stakeholder Consultation information required Provision of an accompanying Guide to help the user understand detailed rules and requirements
1.0	10 July 2017	Initial adoption